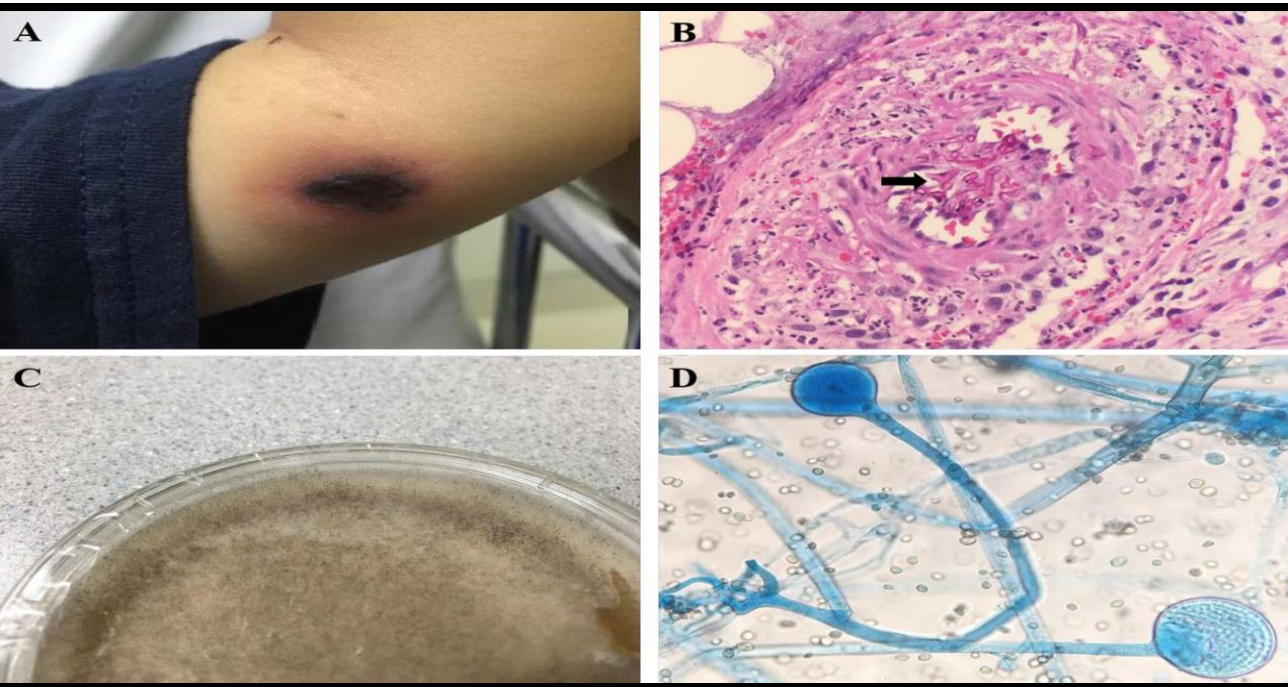


Laboratory Diagnosis of Fungal Infections



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Objectives:

To confirm clinical suspicion to establish fungal cause of disease & clarify the morphology of fungi.

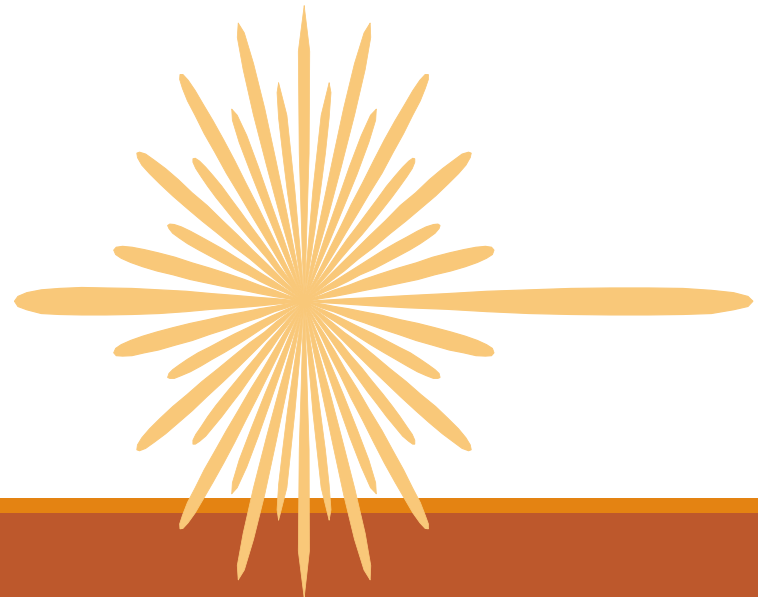
To help in -

- Choosing a therapeutic agent
- Monitoring the course of disease
- Confirming mycological cure

Classification of fungi depending on morphology

Depending on cell morphology

1. Yeasts
2. Molds
3. Yeast like fungi
4. Dimorphic fungi



1. Yeasts

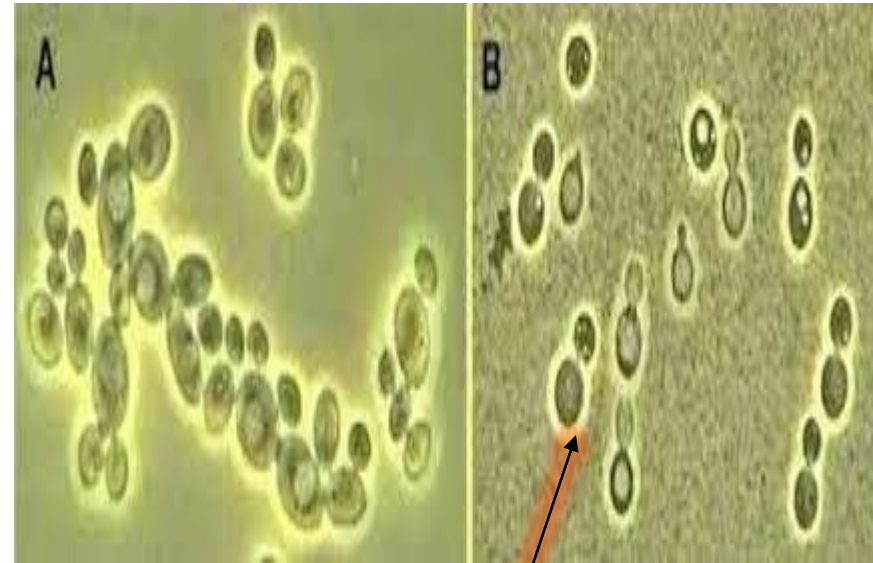
Unicellular fungi, non filamentous, typically oval or spherical cells which reproduce by budding .

► Yeasts are facultative anaerobes, which allows them to grow in a variety of environments.

► *Cryptococcus neoformans* (capsulated yeast)



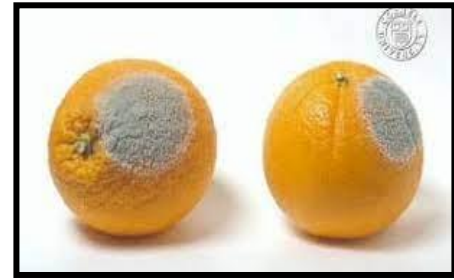
Budding



capsule

2. Molds:

Multicellular, filamentous fungi

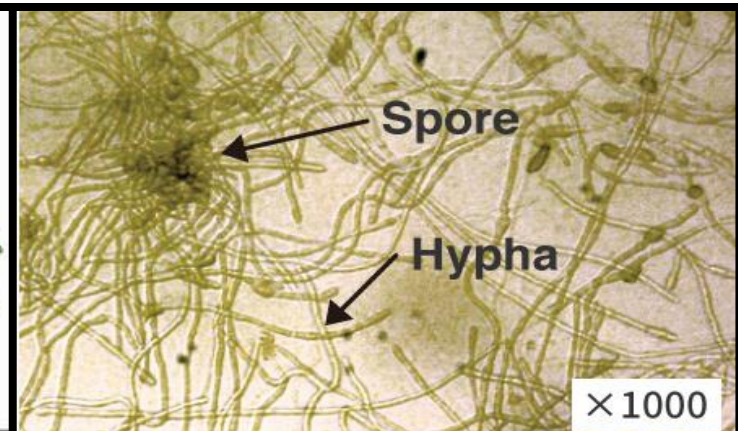


.Hyphae (Sing: Hypha): Long filaments of cells joined together
Cylindrical & branching.

Septate hyphae: regular cross-walls formed in hyphae which divided cell by cross-walls

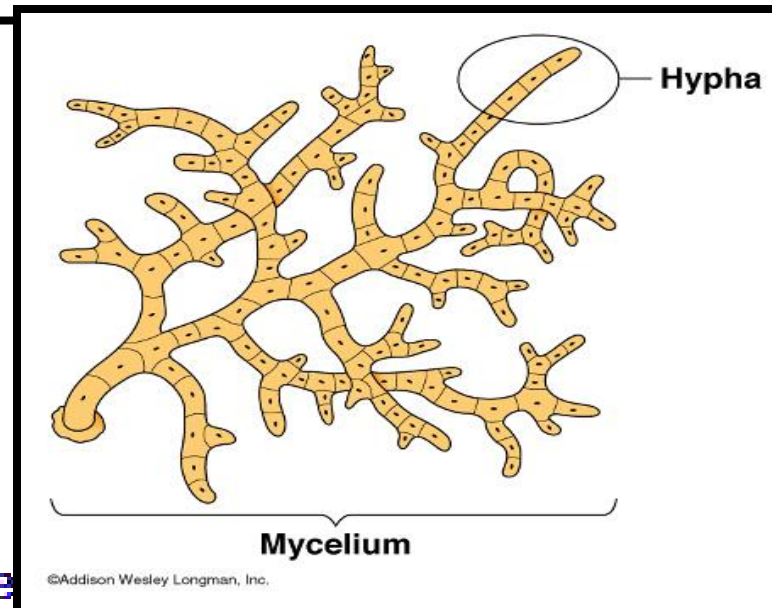
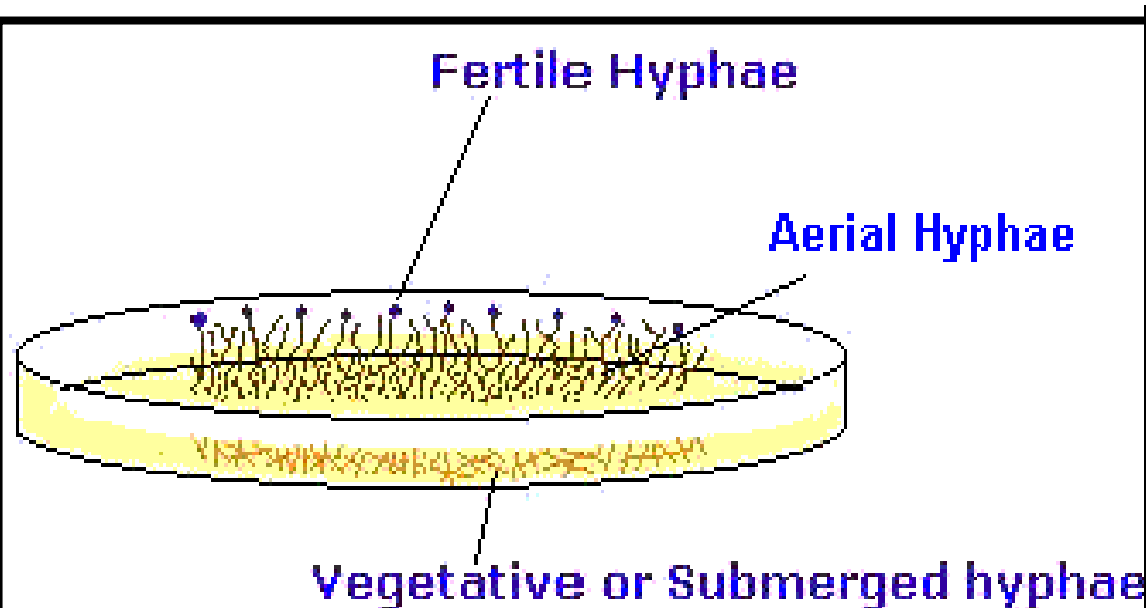
Aseptate hyphae (Coenocytic): Long, continuous cells that are not divided by septa.

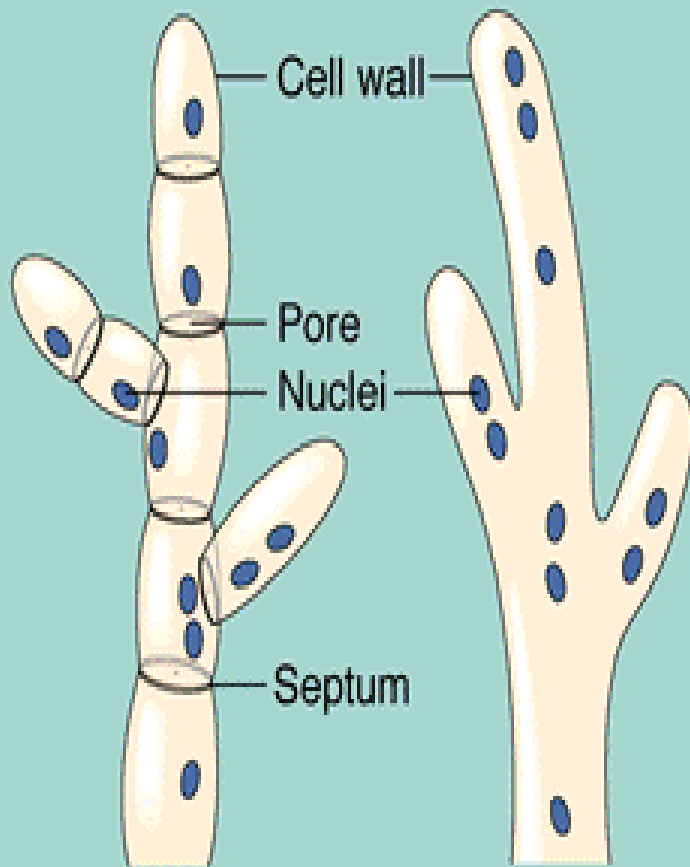
- Hyphae grow by elongating at the tips.
- Each part of a hypha is capable of growth.
- **Mycelium:** Large, visible, filamentous mass made up of many hyphae.



Mycelium are of three kinds in culture:

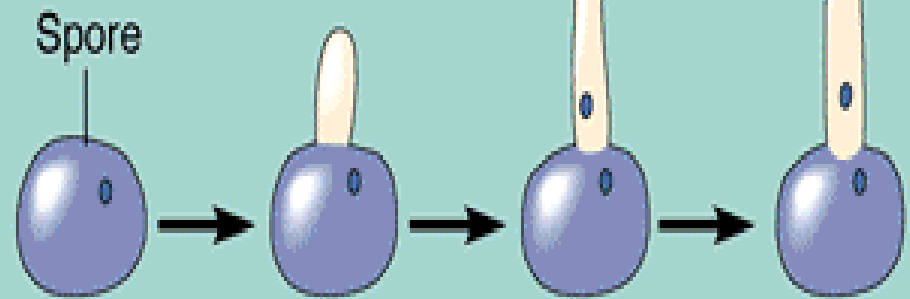
1. Vegetative mycelium are those that penetrates the surface of the medium and absorbs nutrients.
2. Aerial mycelium are those that grow above the agar surface
3. Fertile mycelium are aerial hyphae that bear reproductive structures such as conidia or sporangia.



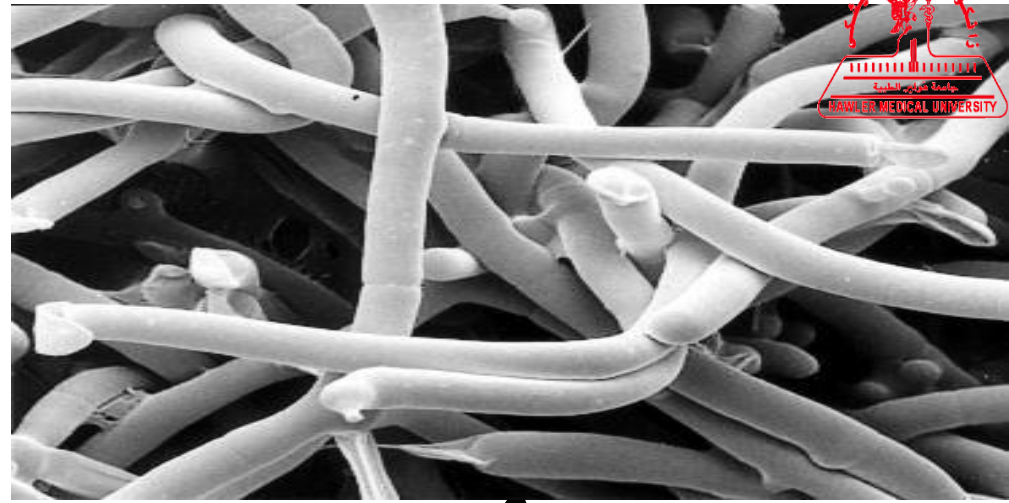
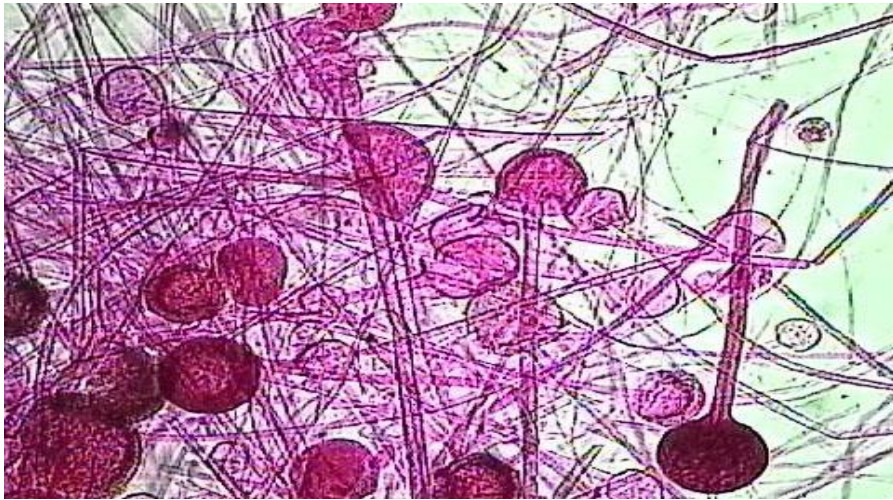


(a) Septate hyphae

(b) Coenocytic hyphae



(c) Growth of a hypha from a spore



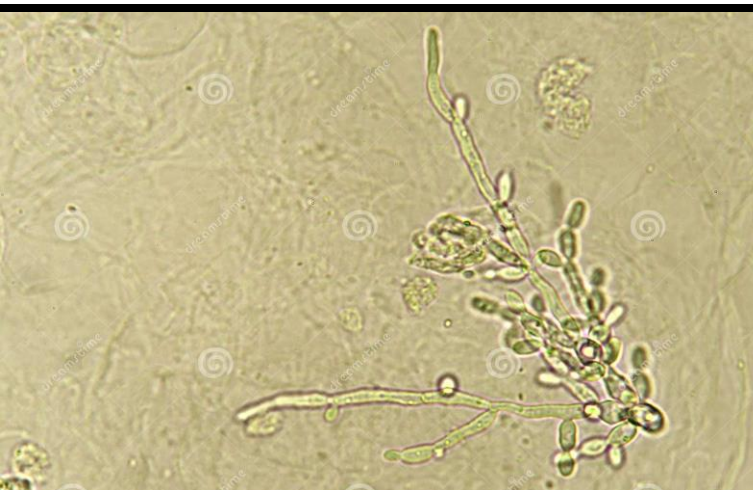
Fungal hypha

Differences Between Mold and Yeast



3- Yeast like fungi

- Grow partly as yeasts and partly as elongated cells resembling hyphae
- ▶ In some cases, as the cells buds the buds fail to detach and elongate thus forming a chain of elongated hyphae like filament called **pseudohyphae**. This property is seen in ***Candida albicans***.



pseudohyphae

4. Dimorphic fungi

Some fungi, including most of the pathogenic fungi exhibit two forms of growth. This is known as **dimorphism**, which exist as mold, filamentous **mycelial** when grown at room temperature **27°C**, and as **yeast** (parasitic phase) inside the body at **37°C**.

- Most fungi causing systemic infections are dimorphic:

- *Histoplasma capsulatum*
- *Blastomyces dermatidis*
- *Paracoccidioides brasiliensis*
- *Coccidioides immitis*
- *Penicillium maneffrei*
- *Sporothrix schenckii*

Thermal Dimorphism in *Sporothrix schenckii*



Sporothrix schenckii - Sabouraud Dextrose Agar, 30°C, 3 weeks incubation

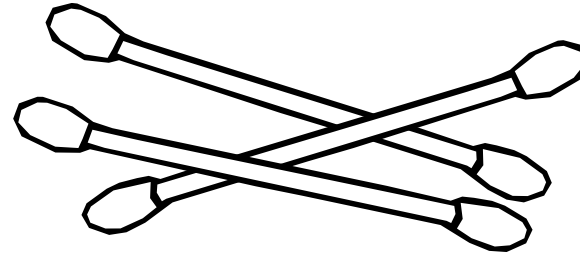
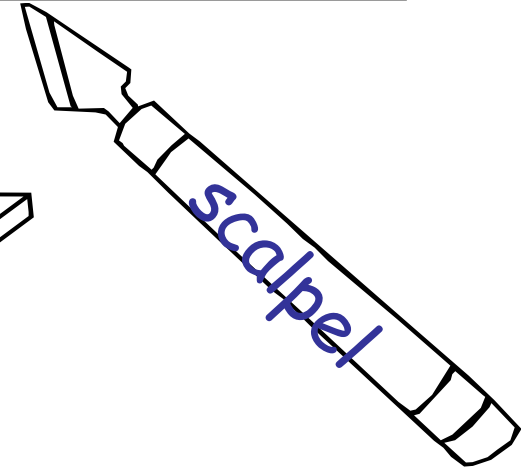
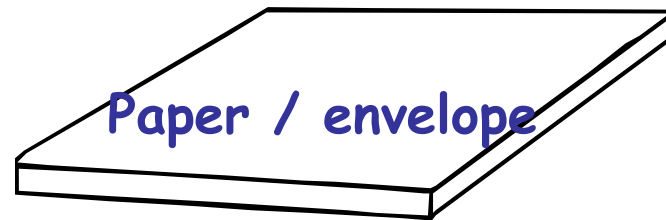


Sporothrix schenckii - Brain-Heart Infusion (BHI) agar, 37°C, 10 days of incubation

Preparation of Specimens for Transport to Laboratory

- Proper collection of specimen and in adequate quantity.
- Early transport to the lab to avoid overgrowth of contaminant.
- Hair & nails sent in a dry envelope, inside proper container.
- Other specimens are usually sent frozen or on dry ice.
- Inside labeling information: Patient ID, specimen source, suspected organism.

Skin scraping specimen



Collection & Transport of specimen

- Specimen collection depends on the corresponding disease.
- Very important to proceed for a final diagnosis.

1- Respiratory specimens

A-Sputum : early morning sample, after mouth wash, flakes to be used for culturing

B-Bronchoscopy : if non productive cough

C-Bronchial brushings or lung biopsy : to rule out invasion or colonisation

2-Blood

- In Brain Heart Infusion agar
- Inoculated in 2 bottles - for dimorphic fungi

3-Cerebrospinal fluid (CSF)

- Should be immediately processed else stored at RT or at 30°C in an incubator
 - Centrifuge & use sediment for culture
-

4-Skin, Hair & Nail

- Taken for dermatophytic infections
- Hair - plucked with forceps

5-Tissue, BM & Body fluids

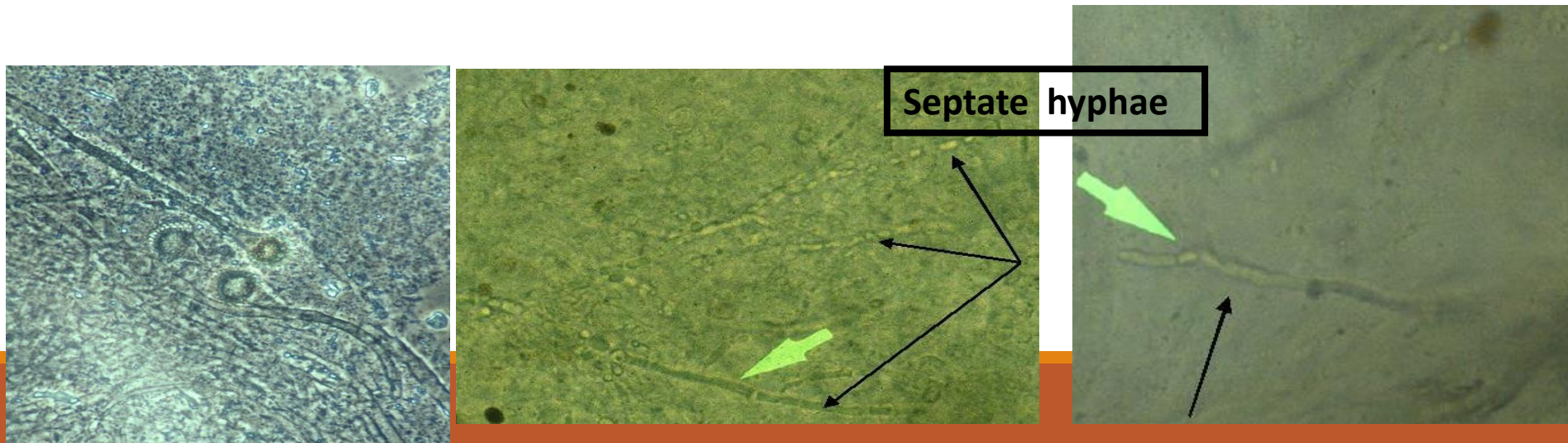
- Tissues - grind or mince before culturing
- Body fluids - centrifuge & use sediment for culture

6-Urine - centrifuge & use sediment for culture

Laboratory Methods for Diagnosis of Mycoses

1. DIRECT EXAMINATION:

10-20% KOH done on skin scrapings, hair, nails, sputum, vaginal specimens, digests protein debris, dissolves keratin, clears the specimen's tissue cells, mucous so fungal elements can be seen.



2. Isolation & Culture

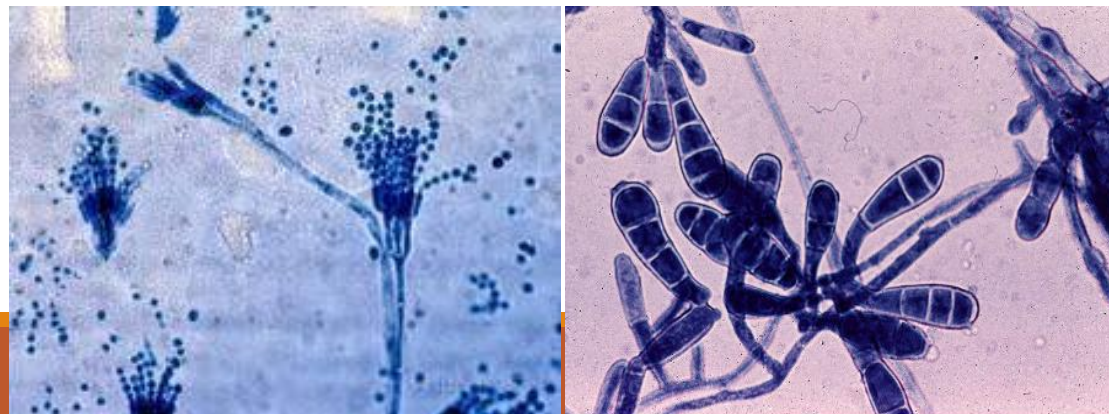
SDA, BHIA

Media with/without antibiotics

Macroscopic examination of culture

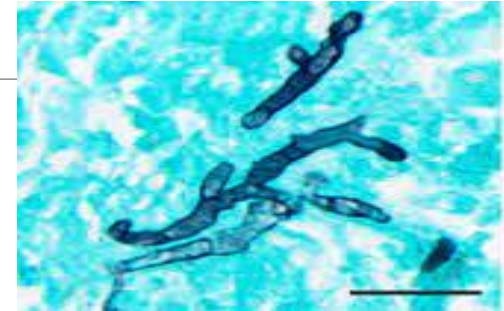
Microscopic examination using LPCB stain by using

Teased Preparations - part of the colony is removed to a slide with inoculating needle. **Lactophenol cotton blue** stain often used to stain the chitin in cell walls of fungi.

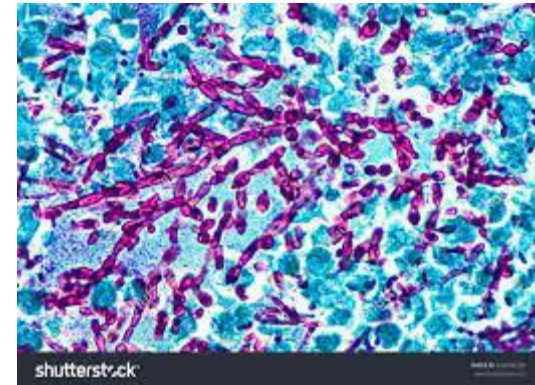


Stains used in Histopathology

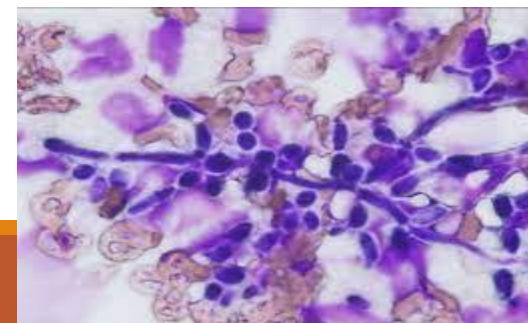
Gomori Methenamine Silver Stain - outlines fungi in black due to the silver precipitating on the fungi cell wall. Internal structures are deep rose to black; background is light green



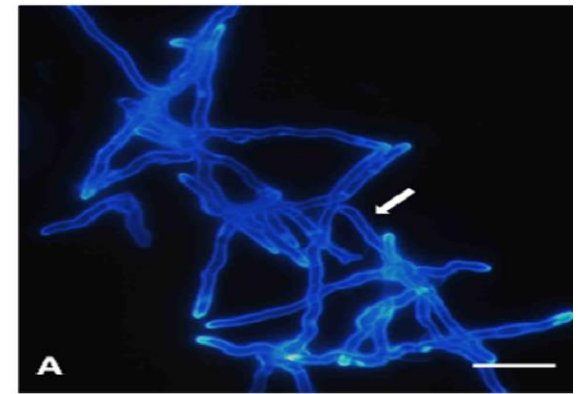
◆ **Per Iodic acid - Schiff Stain (PAS)** - stains certain polysaccharide in the cell walls of fungi. Fungi stain pink-red with blue nuclei



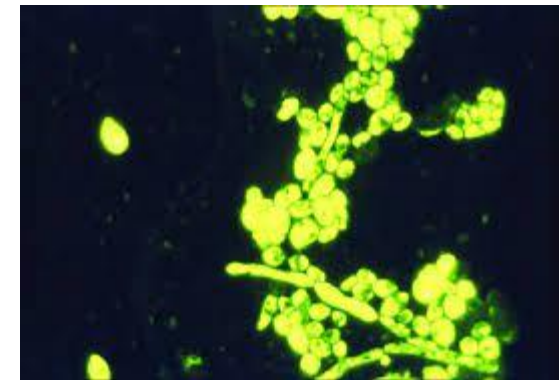
◆ **Giemsa Stain** - use on blood and bone marrow specimens.



- ◆ **Calcofluor white** - fluorescent stain - excellent morphology of pathogenic fungi

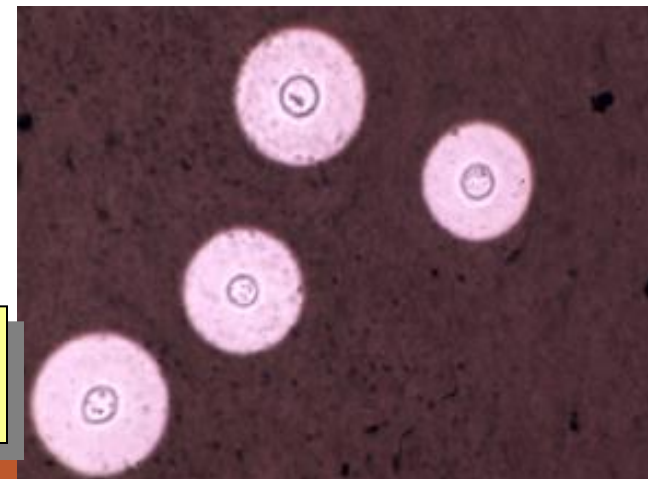


- **Fluorescent Antibody Stain** - simple, sensitive, and specific. To detect **fungal Ag in clinical specimen** such as pus, blood, CSF, tissue sections



- **India Ink** - demonstrates the capsule of *Cryptococcus neoformans* in CSF specimens

India ink -
Cryptococcus



Fungal Culture

-Temperature requirement

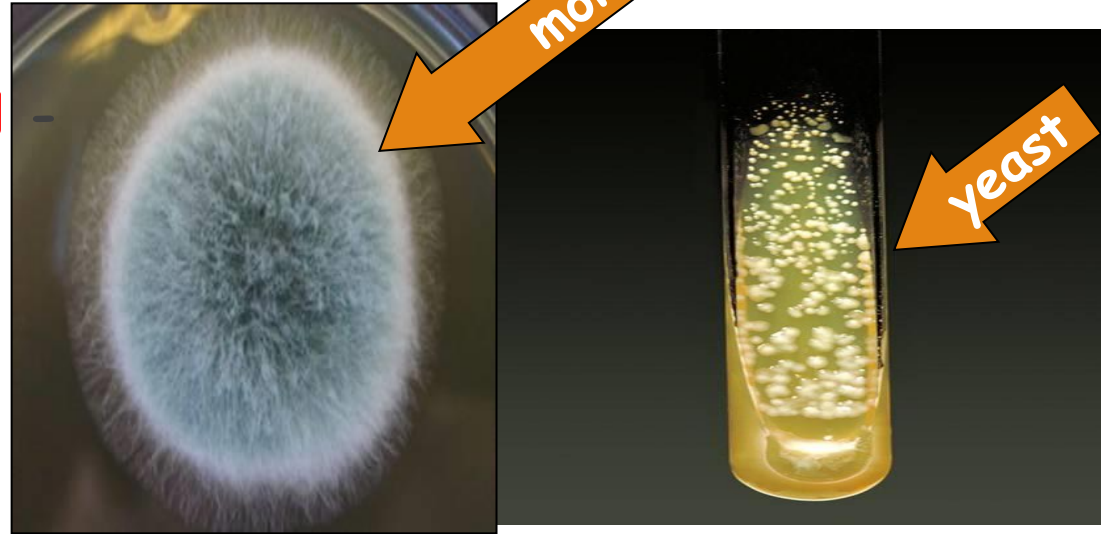
- Majority of fungi - 37°C
- Superficial mycosis - 30°C
- Dimorphic fungi - 25°C & 37°C

- Incubation time

- At least 4 weeks
- Usually positive cultures are obtained in 7-10 days, *Candida* & *Aspergillus* - 24 to 72 hrs

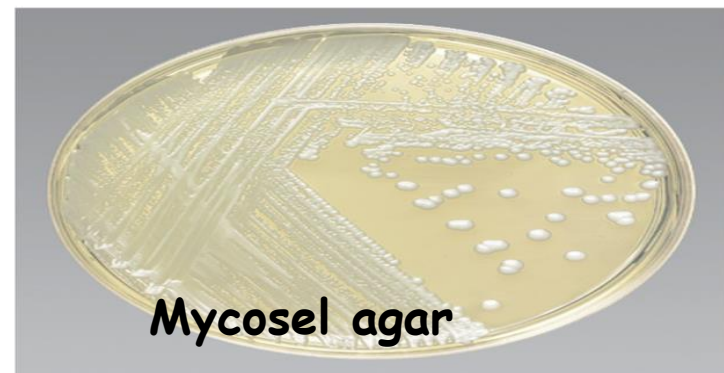
Media Used for Isolation of Fungi

Sabouraud's dextrose agar (SDA) - with chloramphenicol
 chloramphenicol inhibits bacterial growth

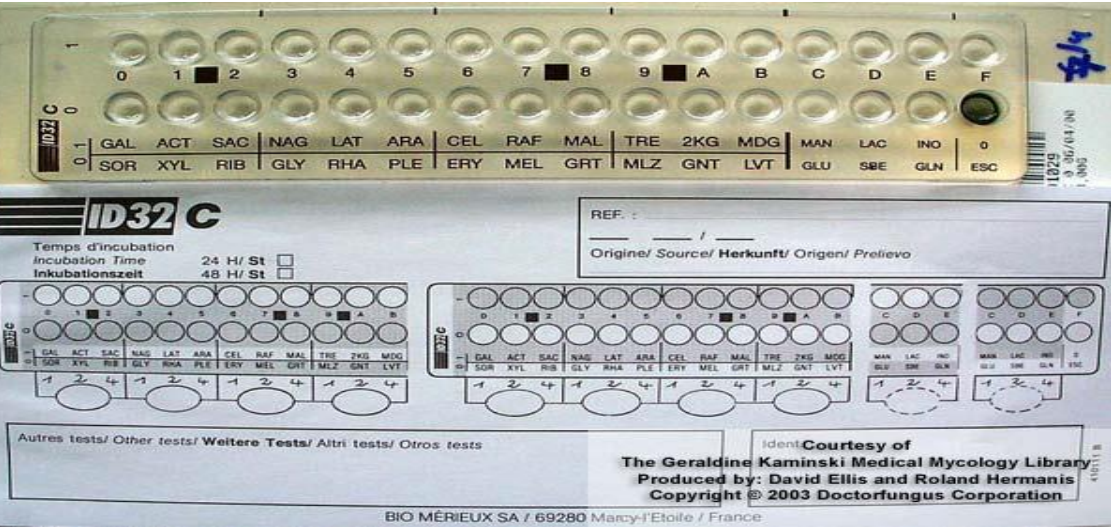


- ◆ Colony morphology - colour, texture, pigment

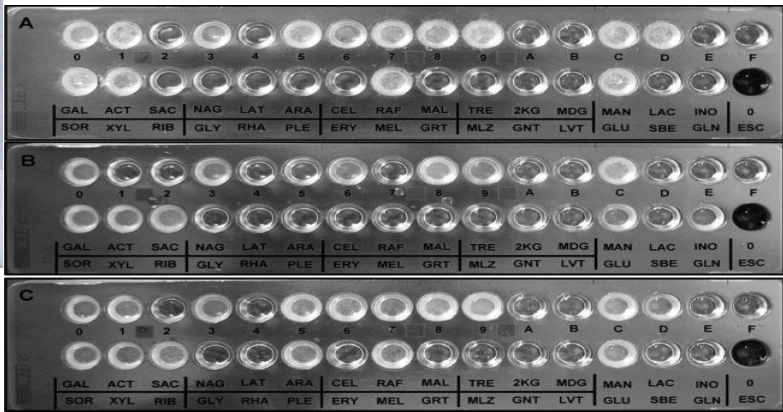
Mycosel agar - Contains chloramphenicol to inhibit bacterial growth; **cycloheximide** to inhibit saprophytic fungi and some yeasts (including *C. neoformans*)



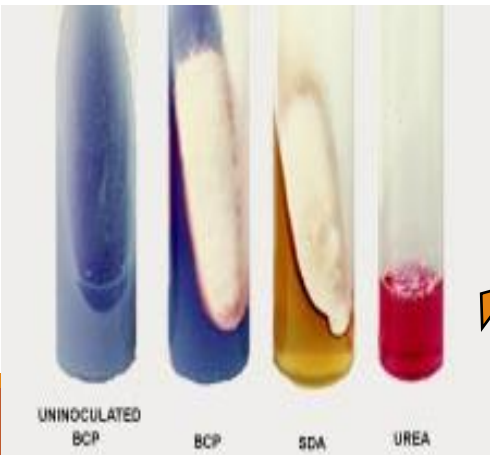
3. Biochemical Tests: Rapid kits for yeasts identification, ability to assimilate carbon & nitrogen, sugar fermentation



Genus/Species: *Candida* species



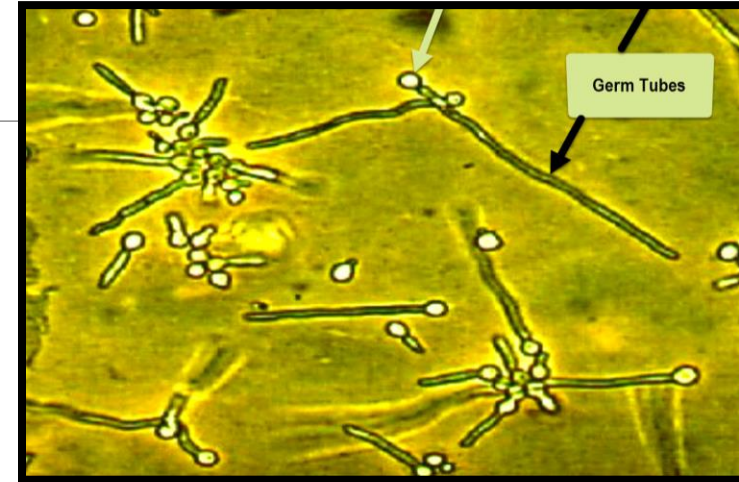
*Urea test



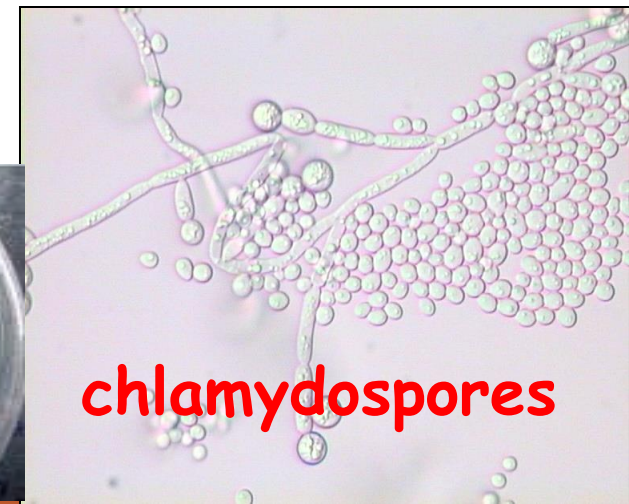
● positive urease production

4. Special Tests:

A-Germ tube - *Candida albicans* & *Candida stellatoidea* produce germ tubes when incubated in a protein medium



B-Demonstration of chlamydospores - yeast is inoculated on (Cornmeal agar with tween 80). Cultures are observed every 24 hours for 3 days looking for chlamydospore production



5-Serological tests - demonstrate B-cell (humoral) immunity to a fungus

6-Skin tests demonstrate T-cell immunity (cellular) to a Fungus

7-PCR & other molecular methods

Skin tests



Thank you 😊

Any question

